

The `measured` package*

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Abstract

The `measured` package provides semantic markup for measurement models. It wraps standard L^AT_EX maths environments (`equation`, `align`) so that metadata about each model equation and its constituent terms are written to a sidecar file (`\jobname.models`) during compilation.

Contents

1	Introduction	1
2	Usage	2
2.1	Model environments	2
2.2	Term definitions	2
2.3	Semantic term macros	2
3	Sidecar file format	2
4	Implementation	2
4.1	Messages	3
4.2	Metadata variables	3
4.3	Output stream	3
4.4	Metadata keys	3
4.5	Model-writing logic	3
4.6	Document environments	4
4.7	Term definition command	5
4.8	Semantic term macros	5

1 Introduction

When writing measurement-model documents it is useful to capture structured metadata alongside the typeset equations. The `measured` package does this transparently: the author writes equations using familiar environments and the package emits a sidecar file.

*This document corresponds to `measured` v0.1, dated 2026-06-02.

2 Usage

Load the package in the usual way:

```
\usepackage{measured}
```

2.1 Model environments

`mdeq` (*env.*) A wrapper for `equation`. An optional key–value argument accepts the keys `name`, `role`, `source`, and `label`.

```
\begin{mdeq}[name=Ohm's law, label=eq:ohm]
  V = IR
\end{mdeq}
```

`mdalign` (*env.*) A wrapper for `align`, with the same metadata keys.

2.2 Term definitions

`\mddefn` `\mddefn{<label>}{<term>}{<definition>}` typesets `<term>` in inline maths mode and writes a TERM record to the sidecar file. A warning is issued if `<label>` has not been defined by a prior `\label` command.

2.3 Semantic term macros

`\nspace` `\nspace{<text>}` renders a namespace identifier in upright roman type.
`\assoc` `\assoc{<text>}` renders an association identifier in upright roman type.
`\tv` `\tv{<text>}` renders a term value in italic type.
`\qual` `\qual{<text>}` or `\qual{<text>}( <sub> )` renders a qualifier, optionally followed by a parenthesised sub-qualifier, both in upright roman type.
`\mdterm` `\mdterm[<ns>]{<sym>}[<qual>]( <assoc> )` composes a decorated term with optional namespace superscript, qualifier subscript, and association argument.

3 Sidecar file format

The file `\jobname.models` contains blocks delimited by header lines:

`===== MODEL =====` starts a model record, followed by key–value lines for `name`, `role`, `source`, `label`, and `equation`.

`--- TERM ---` starts a term record, followed by `eqref`, `term`, and `definition`.

4 Implementation

```
1 <*package>
2 \NeedsTeXFormat{LaTeX2e}
3 \ProvidesExplPackage {measured} {2026-06-02} {0.1}
4   {Semantic markup for measurement models}
5
6 \RequirePackage { amsmath }
```

4.1 Messages

```
7 \msg_new:nnn { measured } { undefined-label }
8 { mddefn:~label~'#1'~is~not~defined. }
```

4.2 Metadata variables

```
9 \tl_new:N \l_md_name_tl
10 \tl_new:N \l_md_role_tl
11 \tl_new:N \l_md_source_tl
12 \tl_new:N \l_md_label_tl
13 \tl_new:N \l_md_model_tl
```

4.3 Output stream

```
14 \iow_new:N \g_md_iow
15
16 \AtBeginDocument
17 {
18   \iow_open:Nn \g_md_iow { \jobname .models }
19 }
20
21 \AtEndDocument
22 {
23   \iow_close:N \g_md_iow
24 }
```

4.4 Metadata keys

```
25 \keys_define:nn { measured / model }
26 {
27   name .tl_set:N = \l_md_name_tl ,
28   role .tl_set:N = \l_md_role_tl ,
29   source .tl_set:N = \l_md_source_tl ,
30   label .tl_set:N = \l_md_label_tl ,
31 }
32
33 \keys_define:nn { measured / mddefn }
34 {
35   model .tl_set:N = \l_md_model_tl ,
36 }
```

4.5 Model-writing logic

`\md_model_write:nn` Clears all metadata keys, applies the supplied key–value pairs, and writes a MODEL block to the sidebar file.

```
37 \cs_new:Npn \md_model_write:nn #1 #2
38 {
39   \tl_clear:N \l_md_name_tl
40   \tl_clear:N \l_md_role_tl
41   \tl_clear:N \l_md_source_tl
42   \tl_clear:N \l_md_label_tl
43
44   \keys_set:nn { measured / model } { #1 }
45
46   \iow_now:Nn \g_md_iow { ===== ~ MODEL ~ ===== }
47 }
```

```

48 \tl_if_empty:NF \l_md_name_tl
49 {
50   \iow_now:Nx \g_md_iow { name: ~ \tl_use:N \l_md_name_tl }
51 }
52
53 \tl_if_empty:NF \l_md_role_tl
54 {
55   \iow_now:Nx \g_md_iow { role: ~ \tl_use:N \l_md_role_tl }
56 }
57
58 \tl_if_empty:NF \l_md_source_tl
59 {
60   \iow_now:Nx \g_md_iow { source: ~ \tl_use:N \l_md_source_tl }
61 }
62
63 \tl_if_empty:NF \l_md_label_tl
64 {
65   \iow_now:Nx \g_md_iow { label: ~ \tl_use:N \l_md_label_tl }
66 }
67
68 \iow_now:Nx \g_md_iow { equation: ~ \tl_to_str:n {#2} }
69 \iow_now:Nn \g_md_iow { }
70 }

```

(End of definition for `\md_model_write:nn`. This function is documented on page ??.)

4.6 Document environments

`mdeq`

```

71 \NewDocumentEnvironment { mdeq } { 0{} +b }
72 {
73   \md_model_write:nn {#1} {#2}
74   \begin{equation}
75     #2
76     \tl_if_empty:NF \l_md_label_tl
77     {
78       \label { \tl_use:N \l_md_label_tl }
79     }
80   \end{equation}
81 }
82 { }

```

(End of definition for `mdeq`. This function is documented on page ??.)

`mdalign`

```

83 \NewDocumentEnvironment { mdalign } { 0{} +b }
84 {
85   \md_model_write:nn {#1} {#2}
86   \begin{align}
87     #2
88     \tl_if_empty:NF \l_md_label_tl
89     {
90       \label { \tl_use:N \l_md_label_tl }
91     }

```

```

92   \end{align}
93   }
94   { }

```

(End of definition for `mdalign`. This function is documented on page ??.)

4.7 Term definition command

`\mddefn` #1 is an equation label (existence is checked), #2 is the term content (typeset in maths mode), #3 is the definition text (sidecar only).

```

95 \NewDocumentCommand \mddefn { m m m }
96 {
97   \cs_if_exist:cF { r@#1 }
98   {
99     \msg_warning:nnn { measured } { undefined-label } { #1 }
100   }
101   \iow_now:Nn \g_md_iow { ----- ~ TERM ~ ----- }
102   \iow_now:Nx \g_md_iow { eqref: ~ \tl_to_str:n { #1 } }
103   \iow_now:Nx \g_md_iow { term: ~ \tl_to_str:n { #2 } }
104   \iow_now:Nx \g_md_iow { definition: ~ \tl_to_str:n { #3 } }
105   \iow_now:Nn \g_md_iow { }
106   $$#2$
107 }

```

(End of definition for `\mddefn`. This function is documented on page ??.)

4.8 Semantic term macros

These use conventional L^AT_EX 2_ε syntax, so we leave `expl3` catcodes temporarily.

```

108 \ExplSyntaxOff

```

`\nspace`

```

109 \NewDocumentCommand{\nspace}{m}{%
110   \mathrm{#1}%
111 }

```

(End of definition for `\nspace`. This function is documented on page ??.)

`\assoc`

```

112 \NewDocumentCommand{\assoc}{m}{%
113   \mathrm{#1}%
114 }

```

(End of definition for `\assoc`. This function is documented on page ??.)

`\tv`

```

115 \NewDocumentCommand{\tv}{m}{%
116   \mathit{#1}%
117 }

```

(End of definition for `\tv`. This function is documented on page ??.)

`\qual`

```
118 \NewDocumentCommand{\qual}{m d()}{%
119   \mathrm{#1}\IfValueT{#2}{(\mathrm{#2})}%
120 }
```

(End of definition for `\qual`. This function is documented on page ??.)

`\mdterm`

```
121 \NewDocumentCommand{\mdterm}{o m o d()}{%
122   \IfValueT{#1}%
123     {\}\nospac{#1}}}%
124   {#2}%
125   \IfValueT{#3}%
126     {\qual{#3}}}%
127   \IfValueT{#4}%
128     {\assoc{#4}}}%
129 }
```

(End of definition for `\mdterm`. This function is documented on page ??.)

```
130 \ExplSyntaxOn
```

```
131 \endinput
```

```
132 \</package>
```

Change History

v0.1

General: Initial public version. 1

Index

The italic numbers denote the pages where the corresponding entry is described, numbers underlined point to the definition, all others indicate the places where it is used.

A		E	
<code>\assoc</code>	2	<code>\end</code>	80, 92
<code>\assoc</code>	<u>112</u> , 128	<code>\endinput</code>	131
<code>\AtBeginDocument</code>	16	environments:	
<code>\AtEndDocument</code>	21	<code>mdalign</code>	2
		<code>mdeq</code>	2
B		<code>\ExplSyntaxOff</code>	108
<code>\begin</code>	74, 86	<code>\ExplSyntaxOn</code>	130
C		I	
cs commands:		<code>\IfValueT</code>	119, 122, 125, 127
<code>\cs_if_exist:NTF</code>	97	iow commands:	
<code>\cs_new:Npn</code>	37	<code>\iow_close:N</code>	23

<code>\iow_new:N</code>	14	<code>mdeq</code>	<u>71</u>
<code>\iow_now:Nn</code>	46, 50, 55,	<code>\mdterm</code>	2
60, 65, 68, 69, 101, 102, 103, 104, 105		<code>\mdterm</code>	<u>121</u>
<code>\iow_open:Nn</code>	18	msg commands:	
		<code>\msg_new:nnn</code>	7
		<code>\msg_warning:nnn</code>	99
J			
<code>\jobname</code>	1, 2, 18	N	
K			
keys commands:		<code>\NeedsTeXFormat</code>	2
<code>\keys_define:nn</code>	25, 33	<code>\NewDocumentCommand</code>	
<code>\keys_set:nn</code>	44		95, 109, 112, 115, 118, 121
L			
<code>\label</code>	78, 90	<code>\NewDocumentEnvironment</code>	71, 83
M			
<code>\mathit</code>	116	<code>\nspace</code>	2
<code>\mathrm</code>	110, 113, 119	<code>\nspace</code>	<u>109</u> , 123
md commands:		P	
<code>\g_md_iow</code> ...	14, 18, 23, 46, 50, 55,	<code>\ProvidesExplPackage</code>	3
60, 65, 68, 69, 101, 102, 103, 104, 105		Q	
<code>\l_md_label_tl</code>		<code>\qual</code>	2
.....	12, 30, 42, 63, 65, 76, 78, 88, 90	<code>\qual</code>	<u>118</u> , 126
<code>\l_md_model_tl</code>	13, 35	R	
<code>\md_model_write:nn</code> ...	37, 37, 73, 85	<code>\RequirePackage</code>	6
<code>\l_md_name_tl</code>	9, 27, 39, 48, 50	T	
<code>\l_md_role_tl</code>	10, 28, 40, 53, 55	tl commands:	
<code>\l_md_source_tl</code> ...	11, 29, 41, 58, 60	<code>\tl_clear:N</code>	39, 40, 41, 42
<code>mdalign (env.)</code>	2	<code>\tl_if_empty:NTF</code> .	48, 53, 58, 63, 76, 88
<code>mdalign</code>	<u>83</u>	<code>\tl_new:N</code>	9, 10, 11, 12, 13
<code>\mddefn</code>	2	<code>\tl_to_str:n</code>	68, 102, 103, 104
<code>\mddefn</code>	<u>95</u>	<code>\tl_use:N</code>	50, 55, 60, 65, 78, 90
<code>mdeq (env.)</code>	2	<code>\tv</code>	2
		<code>\tv</code>	<u>115</u>